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AI-Based Interactive System in Human-Computer Interaction Using Natural Language Processing for Disabled People

Course: 6COM2017 Artificial Intelligence Project (Part 2)

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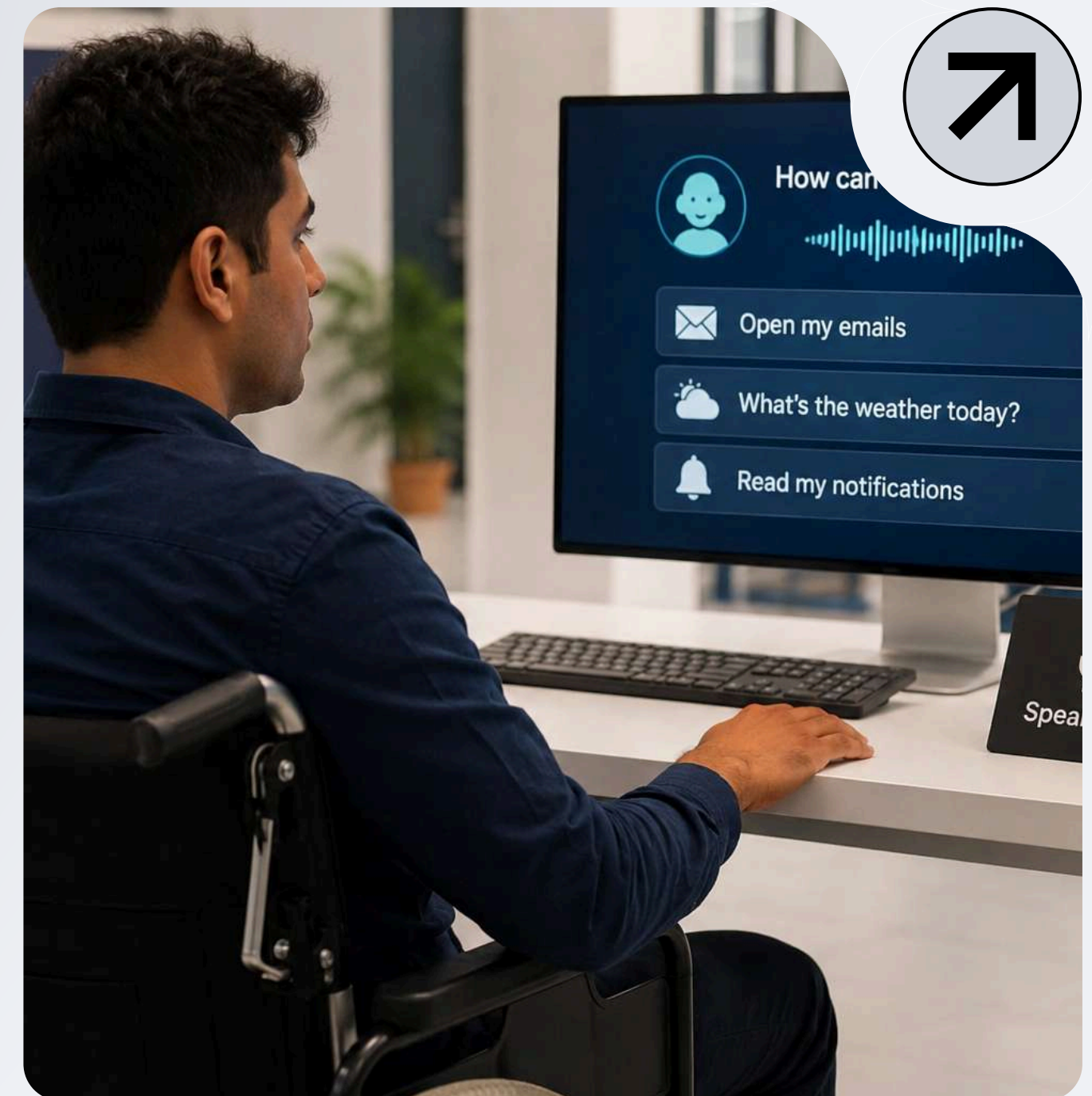
Introduction

Project Overview

- Traditional computer interaction depends on keyboard and mouse devices.
- People with physical disabilities and mobility impairments may face difficulties using these devices.
- Speech Recognition technology enables hands-free interaction.
- Modern systems often depend on internet connectivity and cloud processing.
- This project develops an offline voice-controlled desktop automation system.

Technologies Used

- Artificial Intelligence (AI)
- Natural Language Processing (NLP)
- Automatic Speech Recognition (ASR)
- Desktop Automation



Background & Problem Statement

Background

- Human-Computer Interaction (HCI) is evolving toward natural interfaces.
- Voice interaction provides a more accessible way to control computers.
- Speech recognition is widely used in:
 - Virtual Assistants
 - Healthcare
 - Education
 - Accessibility Systems

Problem Statement

- Traditional input devices limit accessibility.
- Cloud-based speech systems require internet access.
- Privacy and security concerns exist.
- Environmental noise affects recognition accuracy.
- Existing research focuses more on transcription than desktop automation.



AIM

To develop and evaluate an offline voice-controlled desktop automation system capable of performing real-time computer operations.

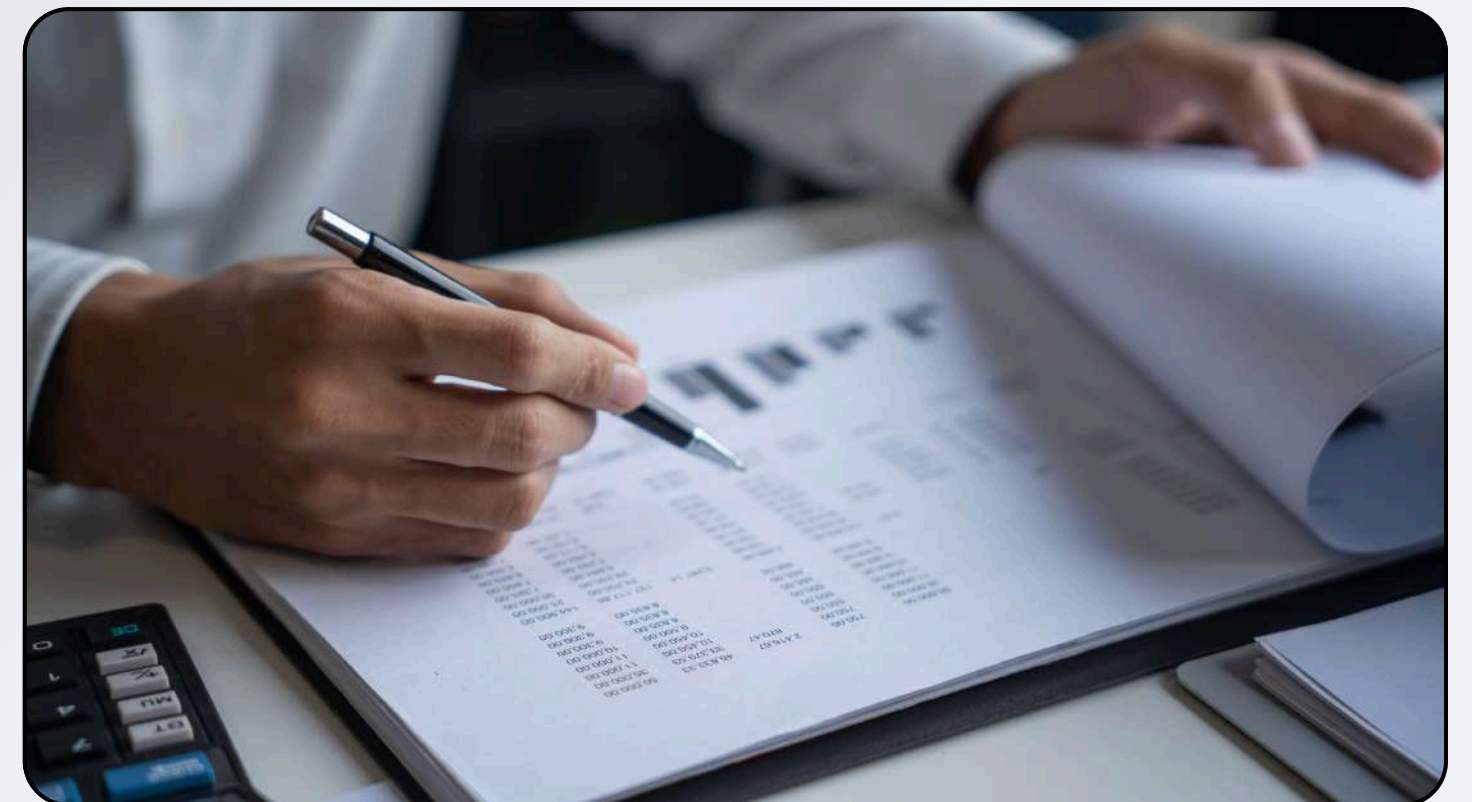
OBJECTIVES

- Develop an offline voice-controlled system.
- Implement desktop automation commands.
- Evaluate recognition accuracy and reliability.
- Analyze environmental noise effects.

RESEARCH QUESTIONS

- How can offline speech recognition be integrated with desktop automation?
- How effective is voice command execution?
- What accuracy and usability levels are achieved?
- How does noise influence performance?

Aim, Objectives & Research Questions



Literature Review

KEY CONCEPTS

Artificial Intelligence (AI)

- Machine learning
- Decision making
- Pattern recognition

Human-Computer Interaction (HCI)

- Improves communication between users and computers
- Supports accessibility

Voice-Based Interaction

- Hands-free operation
- Improved accessibility
- Faster task execution
- Increased user independence

Speech Recognition Technologies

Frameworks Evaluated

Framework	Strength
PocketSphinx	Lightweight & Offline
CMUSphinx	Customizable
Vosk	High Accuracy & Low Latency
Google Speech-to-Text	Cloud-Based Accuracy

- Offline operation
- Low response time
- Better recognition accuracy
- Privacy protection
- Robust performance in noisy environments

Why Vosk?

Methodology

Development Approach

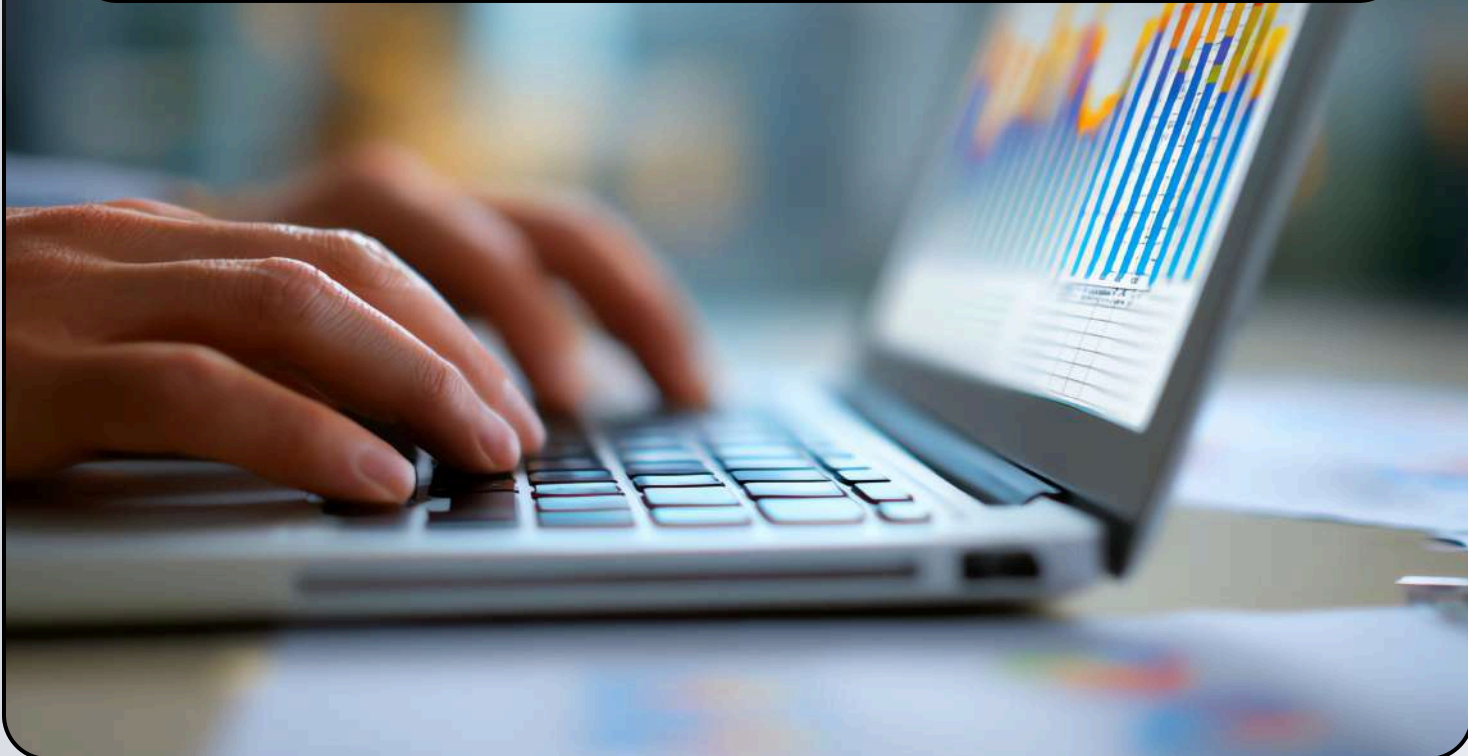
Prototype Development Methodology

Stages:

1. Requirement Analysis
2. System Design
3. Prototype Development
4. Testing
5. Evaluation
6. Refinement

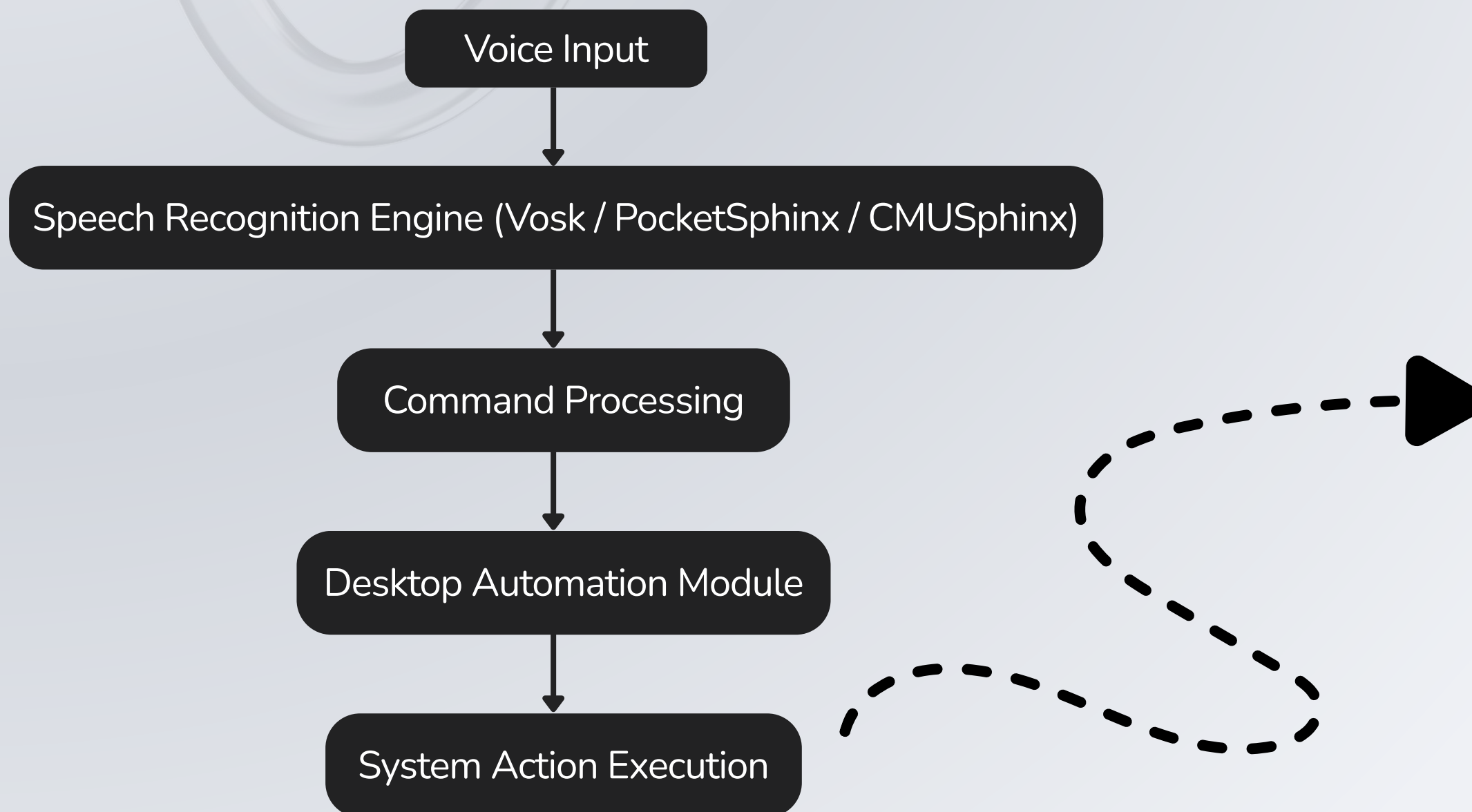
Development Tools

- Python
- Vosk
- PocketSphinx
- CMUSphinx
- PyAutoGUI
- Webbrowser
- Subprocess



Proposed System Architecture

SYSTEM WORKFLOW



Supported Commands

- Mouse movement
- Left click
- Right click
- Double click
- Scroll up/down
- Open browser
- Open YouTube
- Open Calculator
- Open Camera
- Volume control



Implementation & Testing

Evaluation Metrics

- Recognition Accuracy
- Word Error Rate (WER)
- Command Detection Rate
- Runtime Monitoring
- Environmental Noise Testing

Functional Testing

The system successfully performed:

- ✓ Browser Automation
- ✓ Calculator Automation
- ✓ Camera Control
- ✓ Volume Control
- ✓ Mouse Navigation



Results & Discussion

PERFORMANCE COMPARISON

Framework	Performance
Vosk	Best Overall
PocketSphinx	Moderate
CMUSphinx	Moderate
Cloud-Based ASR	High Accuracy but Internet Dependent

Key Findings

- Vosk achieved the best balance between:
 - Accuracy
 - Response Time
 - Reliability
 - Noise Robustness

User Evaluation

- Positive usability feedback
- Successful command execution
- Improved accessibility and hands-free operation

Conclusion & Future Work

Conclusion

The proposed system successfully developed an offline voice-controlled desktop automation solution that enables hands-free computer interaction through speech commands. By integrating NLP and speech recognition technologies, the system improves accessibility for disabled users while ensuring privacy through local processing.

Future Work

- Multilingual speech recognition.
- AI conversational assistant integration.
- Speaker authentication.
- Smart-home integration.
- Deep learning model enhancement.





Thank You